Supplementary material: Retention-constrained class expansion with reference-anchored candidate tables

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Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 4.5. All experiments share the same frozen representations, true class partitions and disjoint support pools.

Appendix A.1. Readouts and evaluation measures

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced k-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the base classifier and calibration protocol with reference-anchored fusion. This comparison changes the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy A_B, macro novel accuracy A_N, all-class macro accuracy, and the harmonic mean in Eq. (14). Both datasets use equal numbers of base and novel

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

Readout	Base	Novel	All	Harmonic
CLIP text	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Visual prototypes	77.53 ± 2.03	60.92 ± 3.94	69.23 ± 2.09	68.17 ± 2.49
Text + prototypes	82.47 ± 0.87	77.53 ± 0.47	80.00 ± 0.56	79.92 ± 0.55
Tip-Adapter (tuned)	78.38 ± 0.24	77.42 ± 1.72	77.90 ± 0.83	77.89 ± 0.83
Raw evidence fusion	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Candidate-normalized table	85.36 ± 0.66	70.83 ± 4.33	78.09 ± 2.02	77.36 ± 2.48
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

Appendix A.2. Recognition after five-shot class expansion

Reference-anchored fusion reaches 86.03 ± 1.40 base accuracy and 66.90 ± 4.23 novel accuracy (Table S1). Its base accuracy is 7.71 percentage points above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches 80.00 ± 0.56 all-class accuracy and 77.53 ± 0.47 novel accuracy. Candidate-normalized fusion reaches 85.36 ± 0.66 base and 70.83 ± 4.33 novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets

On fine-grained Pets expansion, reference anchoring reaches 97.18 ± 2.25 base and 87.79 ± 6.18 novel accuracy (Table S2). Base accuracy is 2.41 points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

Readout	Base	Novel	All	Harmonic
CLIP text	92.87 ± 3.37	94.05 ± 1.35	93.46 ± 1.59	93.43 ± 1.63
Visual prototypes	92.08 ± 2.61	68.30 ± 4.62	80.19 ± 2.33	78.35 ± 2.97
Text + prototypes	94.77 ± 2.91	93.91 ± 1.55	94.34 ± 1.43	94.31 ± 1.44
Tip-Adapter (tuned)	93.56 ± 2.45	94.07 ± 1.18	93.81 ± 1.27	93.80 ± 1.28
Raw evidence fusion	90.13 ± 8.01	91.18 ± 4.27	90.66 ± 6.13	90.63 ± 6.18
Candidate-normalized table	96.85 ± 1.50	86.49 ± 7.64	91.67 ± 4.56	91.29 ± 4.99
Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	92.14 ± 3.23

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

Readout	Base	Novel	All
Text + group offset	78.93 ± 2.75	75.56 ± 4.40	77.24 ± 0.89
Anchored text + prototypes	82.40 ± 2.02	76.19 ± 3.18	79.30 ± 0.59
Frozen RP-ridge (2000)	70.18 ± 0.69	52.52 ± 4.27	61.35 ± 1.83

Oxford-IIIT Pets

Readout	Base	Novel	All
Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31
Anchored text + prototypes	94.58 ± 3.22	90.38 ± 5.41	92.48 ± 1.10
Frozen RP-ridge (2000)	93.49 ± 3.32	73.70 ± 5.13	83.60 ± 1.13

Text–prototype fusion reaches 94.34 ± 1.43 all-class macro accuracy; the reference-anchored table reaches 92.49 ± 2.83 with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only calibration rule.

Appendix A.4. Calibration controls and analytic incremental readouts

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and visual evidence for every class. Harmonic selection searches offsets from -4 to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and 95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

Data	Readout	Base	Novel	All	Feasible
CIFAR	Text + group offset	81.47 ± 1.13	71.49 ± 5.03	76.48 ± 1.99	1/3
CIFAR	Anchored text + prototypes	82.84 ± 1.50	77.11 ± 4.46	79.98 ± 1.61	3/3
CIFAR	Candidate-normalized table	84.69 ± 1.92	72.45 ± 4.14	78.57 ± 1.54	3/3
CIFAR	Reference-anchored table	84.56 ± 1.31	73.88 ± 5.01	79.22 ± 2.21	3/3
Pets	Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31	3/3
Pets	Anchored text + prototypes	94.03 ± 3.37	90.96 ± 5.91	92.50 ± 1.27	3/3
Pets	Candidate-normalized table	97.25 ± 1.78	85.26 ± 7.25	91.26 ± 4.40	3/3
Pets	Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	3/3

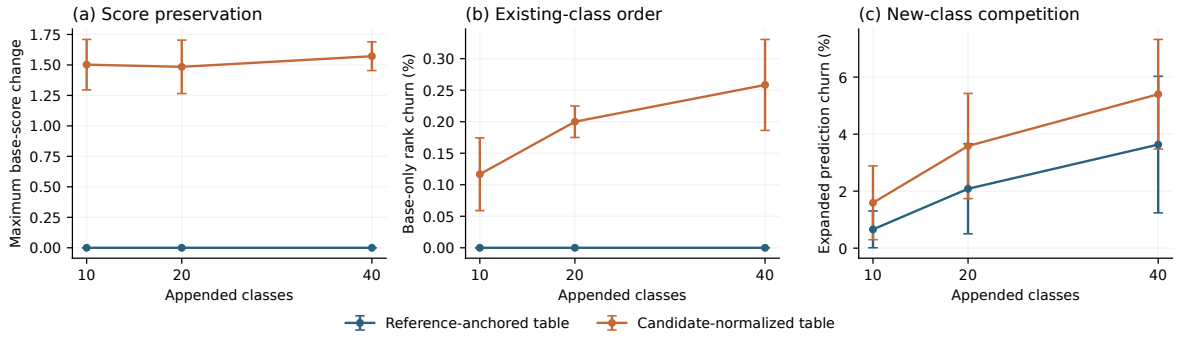


Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

selects a novel offset. This frozen adaptation uses base validation in place of the original within-task MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99% results accompany the experimental records.

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

Shots	Base	Novel	All	Harmonic
1	84.57 ± 0.88	70.86 ± 5.41	77.72 ± 2.27	77.00 ± 2.85
2	85.65 ± 1.45	70.05 ± 1.13	77.85 ± 0.67	77.04 ± 0.67
5	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

Appendix A.5. Vocabulary growth and score preservation

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

Reference anchoring gives zero measured base-score drift and zero base-only ranking churn in all nine CIFAR partition–vocabulary conditions (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of 2.85×10^{-5} on CIFAR and 3.58×10^{-6} on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

Appendix A.6. Support budget and evidence availability

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero branch coefficients, allowing the selected mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

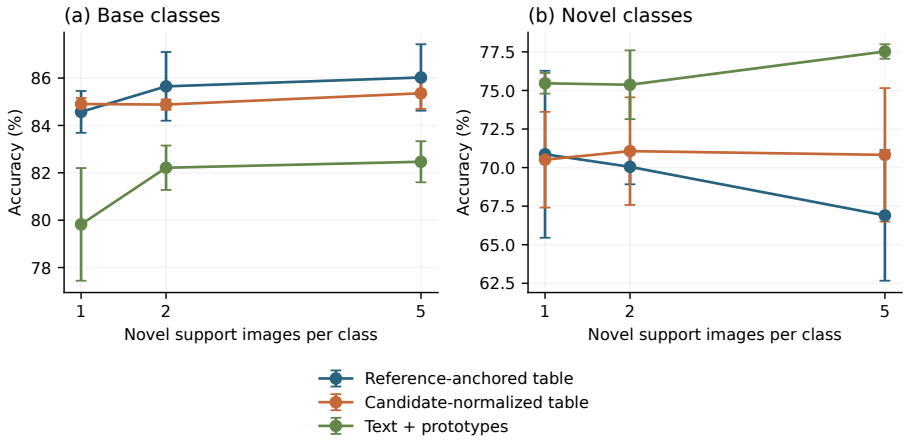


Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

Readout	Base	Novel	All	Harmonic
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14
Without learned evidence	58.10 ± 32.21	80.72 ± 4.47	69.41 ± 13.87	62.85 ± 25.08
Without novel prototypes	86.42 ± 1.71	47.17 ± 30.85	66.79 ± 14.62	56.15 ± 30.05
Without offset	86.14 ± 1.69	65.58 ± 7.35	75.86 ± 2.83	74.25 ± 4.08

Appendix A.7. Mechanism ablation and operating trade-off

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

Figure S3 sweeps d from -2 to 2 in steps of 0.5 with a and u fixed. Increasing d shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (13)). The resulting curve identifies how the offset controls admission of new categories; a validation retention target selects the operating point.

Appendix A.8. Readout efficiency

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization. They include reference statistics and fusion and

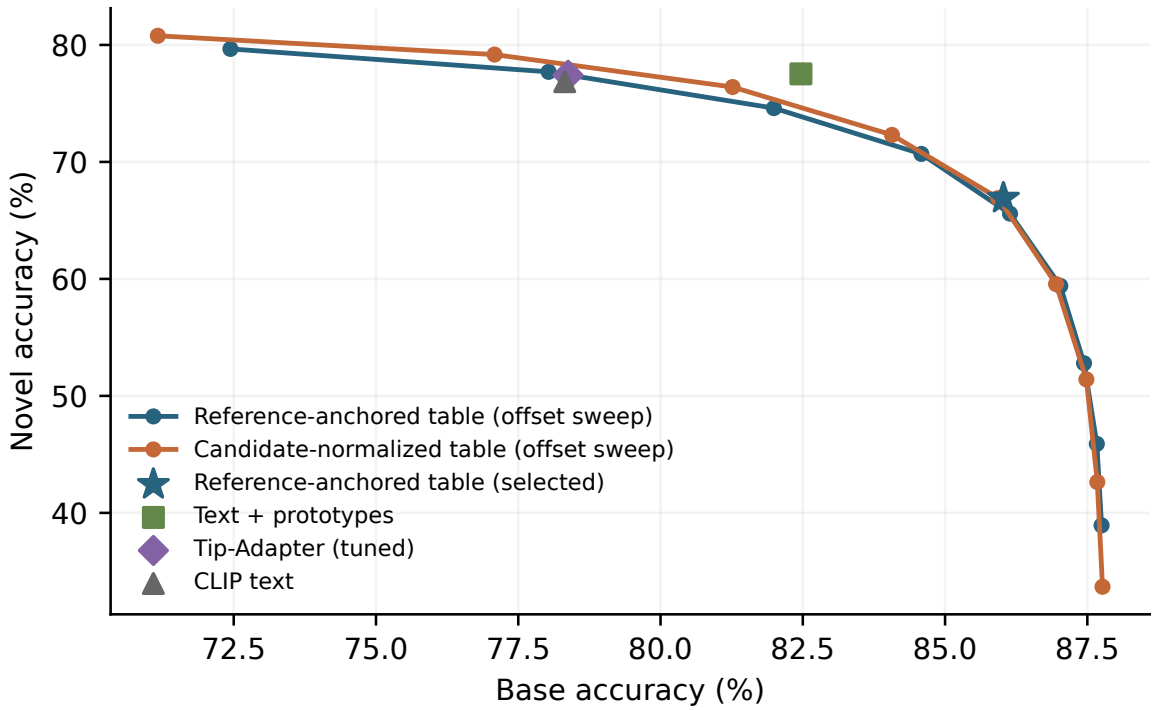


Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR	90%	Retained classifier	78.33 ± 3.89	80.13 ± 2.25	3/3
CIFAR	90%	Classifier removed (fixed gates)	79.07 ± 3.94	70.79 ± 13.81	3/3
CIFAR	90%	No classifier (recalibrated)	78.73 ± 3.21	75.45 ± 5.43	3/3
CIFAR	90%	Anchored text + prototypes	80.18 ± 0.60	79.45 ± 0.76	3/3
CIFAR	95%	Retained classifier	82.99 ± 2.29	75.71 ± 3.14	3/3
CIFAR	95%	Classifier removed (fixed gates)	81.95 ± 0.04	71.54 ± 3.87	2/3
CIFAR	95%	No classifier (recalibrated)	81.85 ± 0.18	71.70 ± 4.10	2/3
CIFAR	95%	Anchored text + prototypes	81.95 ± 0.64	77.48 ± 1.52	2/3
CIFAR	99%	Retained classifier	86.48 ± 1.65	62.83 ± 10.11	3/3
CIFAR	99%	Classifier removed (fixed gates)	–	–	0/3
CIFAR	99%	No classifier (recalibrated)	–	–	0/3
CIFAR	99%	Anchored text + prototypes	86.85	53.85	1/3
Pets	90%	Retained classifier	84.87 ± 4.59	96.53 ± 0.78	3/3
Pets	90%	Classifier removed (fixed gates)	84.11 ± 2.76	96.44 ± 0.64	3/3
Pets	90%	No classifier (recalibrated)	82.00 ± 0.20	96.38 ± 0.52	3/3
Pets	90%	Anchored text + prototypes	85.69 ± 4.62	95.87 ± 1.57	3/3
Pets	95%	Retained classifier	90.29 ± 2.90	95.68 ± 0.83	3/3
Pets	95%	Classifier removed (fixed gates)	90.11 ± 1.17	95.71 ± 0.11	2/3
Pets	95%	No classifier (recalibrated)	89.88 ± 0.85	95.71 ± 0.11	2/3
Pets	95%	Anchored text + prototypes	89.37 ± 4.15	95.63 ± 1.55	3/3
Pets	99%	Retained classifier	96.43 ± 2.15	88.67 ± 7.89	3/3
Pets	99%	Classifier removed (fixed gates)	–	–	0/3
Pets	99%	No classifier (recalibrated)	–	–	0/3
Pets	99%	Anchored text + prototypes	97.70 ± 1.11	84.19 ± 12.46	2/3

Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets

within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are "a photo of a {}.", "a bad photo of a {}.", "a cropped photo of a {}.", "a close-up photo of a {}.", "a bright photo of a {}.", "a good photo of a {}.", "a photo of the {}.", and a second occurrence of "a photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential prototype products can exhibit the reported floating-point residuals when matrix widths change.

Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing d cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase M_N but cannot change M_B. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

Appendix E. Additional background and score-preservation proof

Appendix E.1. Long-tailed recognition and calibration

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pre-trained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

Appendix E.2. Semantic and visual evidence for few-shot recognition

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons

use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

Appendix E.3. Proof of the main score-preservation proposition

Proof. For an existing candidate c, each raw score $q_e(x, c)$ and mask m_{ec} is fixed. The means and variances in Eq. (9) are functions of x and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (10) is unchanged. The coefficients and offset of c are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

Corollary 1. *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

Proof. The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

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